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Short Paper Track

# MERA: Mixture of Experts with Retrieval-Augmented Representation for Modeling Diversified Stock Patterns

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## Introduction

- Most existing methods use the same parameters to fit all samples, without considering that the real stock market often exhibits multiple patterns.
- We propose the Mixture-of-Experts with Retrieval-Augmented repress-entation (MERA) module can capture multiple stock patterns, such as differentiated modeling for high-performing and low-performing stocks.
- To enhance the accuracy of GateNet in distinguishing diverse stock patterns, we utilize self-supervised high-level representations to retrieve similar samples. Both the features and labels of the retrieved samples are strong signals to indicate the target stock pattern.

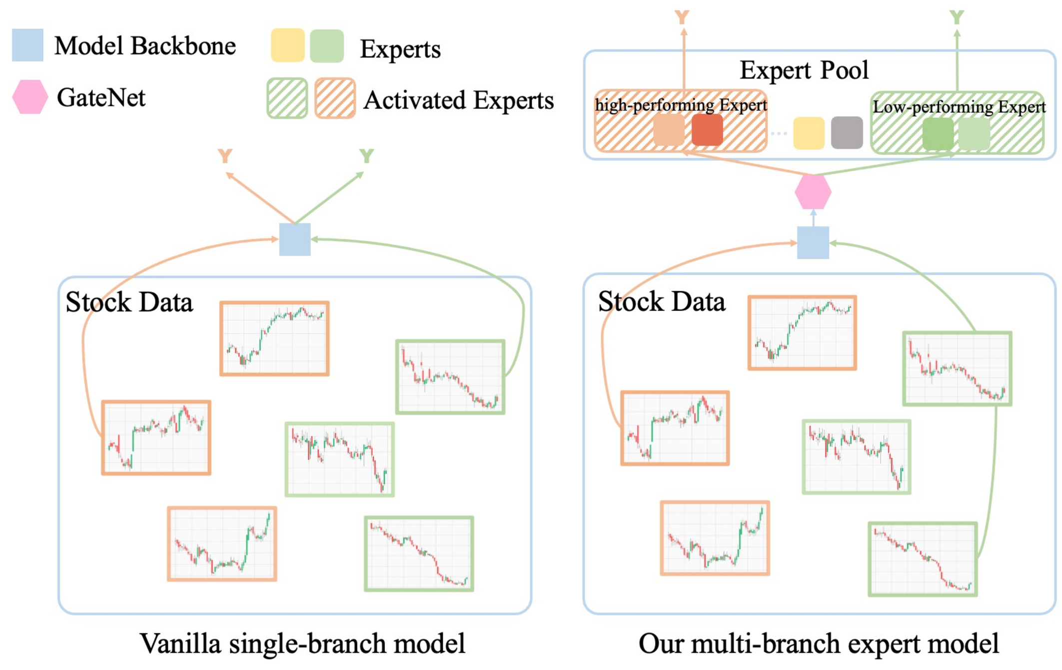


*Figure 1: The comparison between the vanilla single-branch model and our multi-branch expert model.*

## The proposed method

### • Part 1: Self-supervised Pre-training

- Employ a pre-training strategy to transform the raw noisy stock data into compact high-level representations.
- Masked autoencoders: randomly remove portions of the input sequence and train the model to reconstruct the missing parts.
- Store both the pre-trained high-level representations and the label information into the retrieval pool.
- Continuous labels are discretized into categorical variables to indicate stock patterns and determine which expert modules to activate.

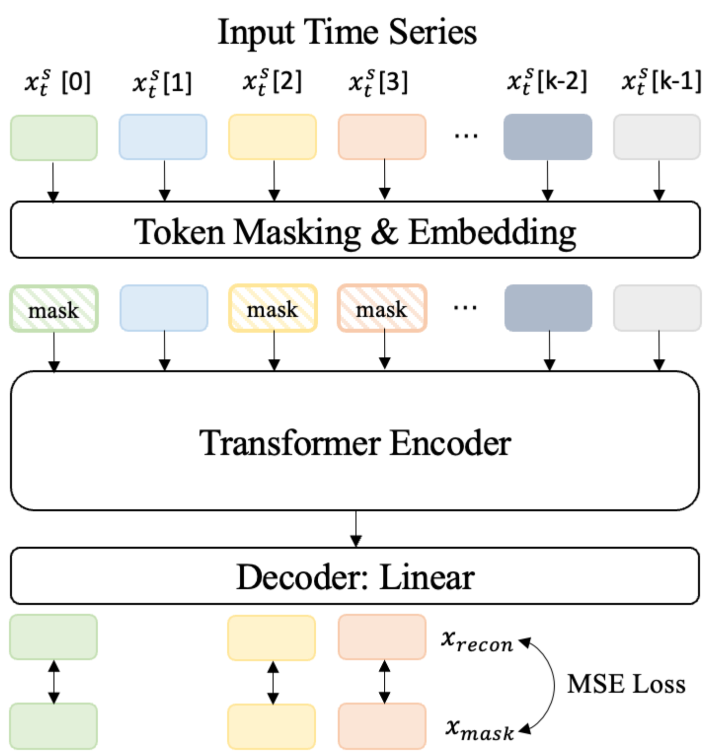


*Figure 2: Masked self-supervised representation learning.*

### • Part 2: Retrieval Augmentation

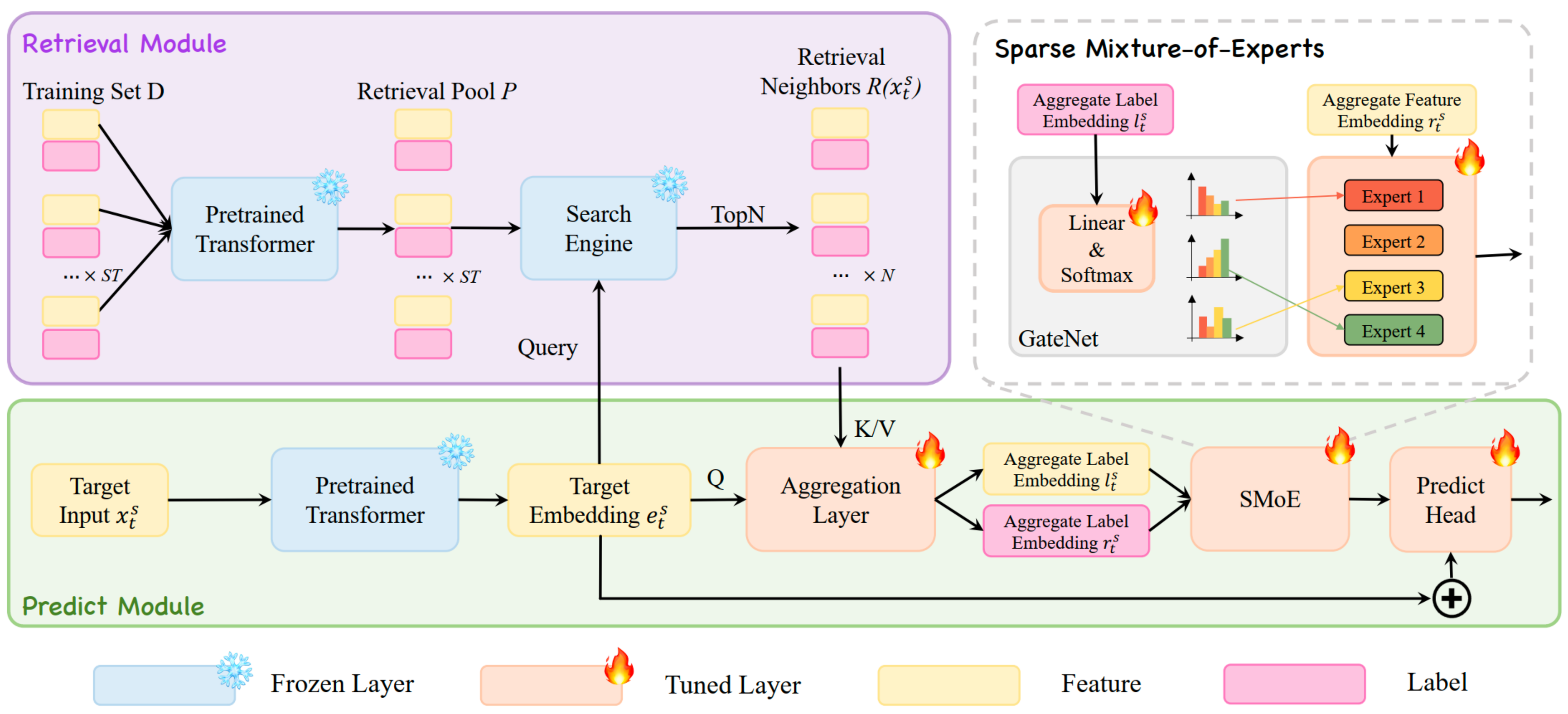


*Figure 3: The overall architecture of our MERA Module.*

- Given a prediction request $x_t^s$, the pre-trained model backbone is first used to extract its feature embedding $e_t^s$, which will be regarded as a query to retrieve $TopN$ similar samples from the retrieval pool.

## The proposed method

- These retrieved samples are then aggregated using a target-aware, order-invariant attention mechanism, which assigns a relevance-based scaling coefficient to each sample.

$$r_t^s = \sum_{i=1}^{N} \alpha_i \cdot s_i^f, s_i \in \mathcal{R}\left(x_t^s\right) \qquad \alpha_i = \frac{\exp\left(\boldsymbol{x}_i^T \boldsymbol{W} \boldsymbol{x}_t^s\right)}{\sum_{j=1}^{N} \exp\left(\boldsymbol{x}_j^T \boldsymbol{W} \boldsymbol{x}_t^s\right)}$$

$$l_t^s = \sum_{i=1}^{N} \alpha_i \cdot s_i^l, s_i \in \mathcal{R}\left(x_t^s\right)$$

- To capture the diversified stock patterns, predict module use the SMoE layer for differentiated modeling. Given stock data, the output is a weighted sum of the outputs from $k$ most relevant experts, with the weights determined by GateNet $G$.
- Aggregated label embedding $l_t^s$ is a strong signal to indicate the target pattern and is sent to GateNet for accurate allocation. The features ($e_t^s$, $r_t^s$) are combined and fed into the expert modules for more fine-grained analysis.

$$G(l_t^s) = TopK(Softmax(l_t^s \cdot W_g)) \qquad \hat{y}_t^s = MLP(\sum_{k=1}^{K} G_k(l_t^s) \cdot E_k((e_t^s, r_t^s)))$$

## Experiments

### • Part 1: Stock prediction performance comparison

*Table 1: Quantitative comparisons among different methods in three dataset.*

| Dataset | Method | Ranking Metrics | | | | Portfolio Metrics | | |
|---|---|---|---|---|---|---|---|---|
| | | IC | ICIR | RankIC | RankIR | *Return*$^+$ | *Return*$^-$ | Return |
| CSI300 | MLP | 0.071 | 0.830 | 0.068 | 0.846 | 0.431 | -0.772 | 0.602 |
| | LightGBM | 0.092 | 0.814 | 0.080 | 0.807 | 0.614 | -0.911 | 0.763 |
| | TCN | 0.087 | 0.732 | 0.085 | 0.830 | 0.543 | -0.974 | 0.759 |
| | SFM | 0.080 | 0.801 | 0.079 | 0.823 | 0.482 | -0.922 | 0.702 |
| | GRU | 0.091 | 0.796 | 0.079 | 0.742 | 0.606 | -0.909 | 0.757 |
| | ALSTM | 0.093 | 0.775 | 0.082 | 0.712 | 0.599 | -0.988 | 0.794 |
| | Transformer | 0.095 | 0.821 | 0.094 | 0.848 | 0.658 | -1.015 | 0.837 |
| | Transformer+TRA | 0.098 | 0.827 | 0.096 | 0.848 | 0.701 | -1.047 | 0.874 |
| | Transformer+CISP | 0.102 | 0.832 | 0.100 | 0.850 | 0.707 | -1.064 | 0.886 |
| | Transformer+MERA | **0.107**↑ 0.012 | **0.855**↑ 0.034 | **0.106**↑ 0.012 | **0.859**↑ 0.011 | **0.726**↑ 0.068 | **-1.104**↑ 0.089 | **0.915** ↑ 0.078 |
| CSI500 | MLP | 0.080 | 0.962 | 0.078 | 1.011 | 0.701 | -1.023 | 0.862 |
| | LightGBM | 0.100 | 0.997 | 0.093 | 0.178 | 0.874 | -1.343 | 1.109 |
| | SFM | 0.089 | 0.888 | 0.092 | 1.203 | 0.799 | -1.369 | 1.084 |
| | TCN | 0.100 | 0.969 | 0.096 | 1.184 | 0.847 | -1.393 | 1.120 |
| | GRU | 0.101 | 0.970 | 0.090 | 1.024 | 0.798 | -1.333 | 1.066 |
| | ALSTM | 0.104 | 0.981 | 0.101 | 1.142 | 0.868 | -1.433 | 1.151 |
| | Transformer | 0.109 | 1.045 | 0.102 | 1.203 | 0.889 | -1.524 | 1.207 |
| | Transformer+TRA | 0.114 | 1.089 | 0.106 | 1.250 | 1.012 | -1.558 | 1.275 |
| | Transformer+CISP | 0.113 | 1.092 | 0.108 | 1.257 | 1.013 | -1.556 | 1.280 |
| | Transformer+MERA | **0.117**↑ 0.008 | **1.097**↑ 0.052 | **0.112**↑ 0.010 | **1.262**↑ 0.059 | **1.018**↑ 0.129 | **-1.551**↑ 0.027 | **1.285** ↑ 0.078 |
| CSI1000 | MLP | 0.093 | 1.211 | 0.086 | 1.241 | 1.171 | -2.113 | 1.643 |
| | LightGBM | 0.120 | 1.365 | 0.093 | 1.152 | 1.498 | -2.414 | 1.596 |
| | SFM | 0.113 | 1.396 | 0.092 | 1.232 | 1.315 | -2.423 | 1.869 |
| | TCN | 0.121 | 1.393 | 0.099 | 1.200 | 1.521 | -2.640 | 2.081 |
| | GRU | 0.119 | 1.411 | 0.097 | 1.132 | 1.504 | -2.538 | 2.022 |
| | ALSTM | 0.122 | 1.371 | 0.090 | 1.017 | 1.479 | -2.620 | 2.050 |
| | Transformer | 0.124 | 1.377 | 0.108 | 1.203 | 1.564 | -2.601 | 2.083 |
| | Transformer+TRA | 0.125 | 1.432 | 0.108 | 1.190 | 1.662 | -2.690 | 2.165 |
| | Transformer+CISP | 0.126 | 1.437 | 0.108 | 1.195 | 1.669 | -2.685 | 2.170 |
| | Transformer+MERA | **0.129**↑ 0.005 | **1.444**↑ 0.067 | **0.110**↑ 0.002 | **1.256**↑ 0.053 | **1.675**↑ 0.111 | **-2.677**↑ 0.076 | **2.176** ↑ 0.093 |

### • Part 2: Visualization of data allocation

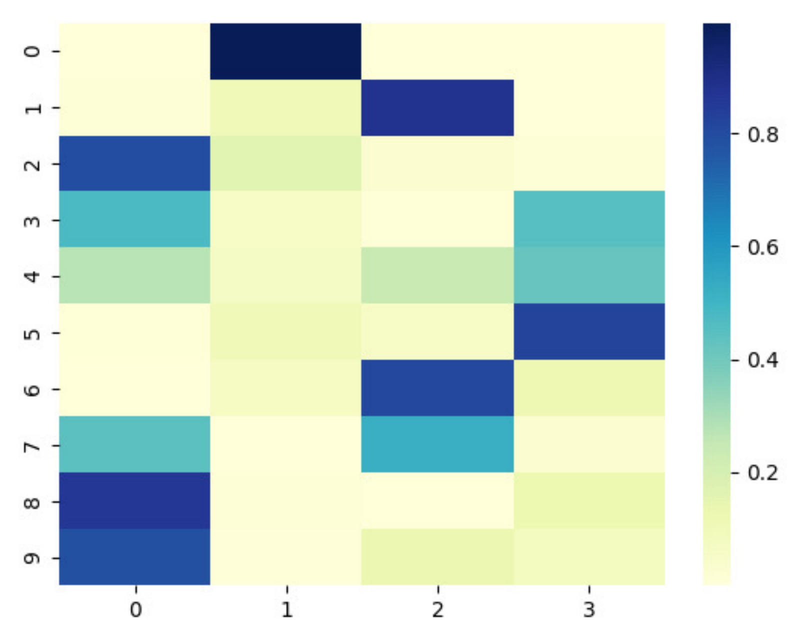


*Figure 4: Visualization of the logits that experts are selected for each label class.*

**Analysis:** To demonstrate that MERA module adaptively allocates data of different patterns to the appropriate experts for analysis, we visualizes the logits of experts being selected for each label class. The activation map exhibits extremely high sparsity. It can be observed that label class 3/4/5 select expert 3, while the label class 7/8/9 select expert 0, indicating that expert allocation is not random, and exhibits a strong correlation with stock patterns.